1 I

1.1 (1)

The Hamiltonian of a one-dimensional classical harmonic oscillator is

$$H = \frac{p^2}{2m} + \frac{m\omega_0^2}{2}u^2.$$

The classical partition function is obtained by integrating over phase space:

$$Z = \frac{1}{h}\int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} du\, e^{-\beta H}, \qquad \beta = \frac{1}{k_B T}.$$

Since the kinetic and potential terms are independent,

$$\begin{aligned} Z &= \frac{1}{h}\int_{-\infty}^{\infty} e^{-\beta p^2/(2m)}\, dp \int_{-\infty}^{\infty} e^{-\beta m\omega_0^2 u^2/2}\, du \\ &= \frac{1}{h}\sqrt{\frac{2\pi m}{\beta}}\sqrt{\frac{2\pi}{\beta m\omega_0^2}} \\ &= \frac{2\pi}{h\beta\omega_0}. \end{aligned}$$

Using $h = 2\pi\hbar$, we obtain

$$\boxed{Z = \frac{1}{\beta\hbar\omega_0} = \frac{k_B T}{\hbar\omega_0}}.$$

Here, the factor $1/h$ makes the classical phase-space integral dimensionless; omitting it only changes Z by a temperature-independent factor and does not affect the energy or heat capacity.

1.2 (2)

The average energy is

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= -\frac{\partial}{\partial \beta}\ln\left(\frac{1}{\beta\hbar\omega_0}\right) \\ &= \frac{1}{\beta} = k_B T. \end{aligned}$$

Therefore, the heat capacity is

$$\boxed{C = \frac{\partial E}{\partial T} = k_B}.$$

Thus, $C/k_B = 1$ for all $k_B T/(\hbar\omega_0)$ in the classical model. Its graph is a horizontal line at $C/k_B = 1$.

1.3 (3)

For a quantum harmonic oscillator, the energy levels are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega_0, \qquad n = 0, 1, 2, \ldots.$$

The partition function is

$$\begin{aligned} Z &= \sum_{n=0}^{\infty} \exp\left[-\beta\left(n+\frac{1}{2}\right)\hbar\omega_0\right] \\ &= e^{-\beta\hbar\omega_0/2}\sum_{n=0}^{\infty} e^{-n\beta\hbar\omega_0} \\ &= \boxed{\frac{e^{-\beta\hbar\omega_0/2}}{1-e^{-\beta\hbar\omega_0}}}. \end{aligned}$$

Taking the logarithm,

$$\ln Z = -\frac{\beta\hbar\omega_0}{2} - \ln\left(1-e^{-\beta\hbar\omega_0}\right).$$

Hence, the average energy is

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial\beta} \\ &= \boxed{\frac{\hbar\omega_0}{2} + \frac{\hbar\omega_0}{e^{\beta\hbar\omega_0}-1}}. \end{aligned}$$

The first term is the zero-point energy.

1.4 (4)

Since the zero-point energy is independent of temperature, it does not contribute to the heat capacity. Thus,

$$\begin{aligned} C &= \frac{\partial E}{\partial T} \\ &= \frac{(\hbar\omega_0)^2}{k_B T^2}\frac{e^{\hbar\omega_0/(k_B T)}}{\left(e^{\hbar\omega_0/(k_B T)}-1\right)^2}. \end{aligned}$$

Let

$$x = \frac{k_B T}{\hbar\omega_0}.$$

Then

$$\boxed{\frac{C}{k_B} = \frac{1}{x^2}\frac{e^{1/x}}{\left(e^{1/x}-1\right)^2}}.$$

At low temperatures ($x \to 0$),

$$\frac{C}{k_B} \sim \frac{e^{-1/x}}{x^2} \longrightarrow 0.$$

At high temperatures ($x \to \infty$),

$$\frac{C}{k_B} \longrightarrow 1.$$

Therefore, the graph of C/k_B increases from zero at low temperatures and approaches the classical limit of 1 at high temperatures.

2 II

2.1 (5)

The potential energy of the one-dimensional chain is

$$V = \frac{m\omega_0^2}{2}\sum_{j=1}^{N}(u_{j+1} - u_j)^2,$$

with periodic boundary conditions $u_{N+1} = u_1$ and $u_0 = u_N$. The force acting on the j-th mass is

$$\begin{aligned} F_j &= -\frac{\partial V}{\partial u_j} \\ &= -m\omega_0^2\left[(u_j - u_{j-1}) + (u_j - u_{j+1})\right] \\ &= m\omega_0^2(u_{j-1} + u_{j+1} - 2u_j). \end{aligned}$$

Therefore, the equation of motion is

$$\boxed{m\ddot{u}_j = m\omega_0^2(u_{j-1} + u_{j+1} - 2u_j)}.$$

2.2 (6)

Assume a normal-mode solution of the form

$$u_j = a_k e^{i(kj - \omega t)}.$$

The periodic boundary condition $u_{j+N} = u_j$ requires

$$e^{ikN} = 1,$$

so

$$\boxed{k = \frac{2\pi n}{N}, \qquad n \in \mathbb{Z}}.$$

Wave numbers differing by integer multiples of 2π describe the same mode. Thus, we choose N inequivalent values in one Brillouin zone, for example $n = 0, 1, \ldots, N-1$. The $k = 0$ mode is a uniform translation, which is excluded as an independent vibrational mode because the center of mass is fixed.

2.3 (7)

Substituting the normal-mode solution into the equation of motion,

$$\begin{aligned} -\omega^2 &= \omega_0^2(e^{-ik} + e^{ik} - 2) \\ &= 2\omega_0^2(\cos k - 1). \end{aligned}$$

Hence,

$$\begin{aligned} \omega^2 &= 2\omega_0^2(1 - \cos k) \\ &= 4\omega_0^2 \sin^2\frac{k}{2}. \end{aligned}$$

Therefore, the dispersion relation is

$$\boxed{\omega(k) = 2\omega_0 \left|\sin\frac{k}{2}\right|}.$$

For $0 < k \ll 1$, using $\sin(k/2) \simeq k/2$, we obtain

$$\boxed{\omega(k) \simeq \omega_0 k}.$$

More generally, near $k = 0$, $\omega(k) \simeq \omega_0|k|$.

2.4 (8)

Each nonzero normal mode can be treated as an independent quantum harmonic oscillator. Its partition function is

$$Z_k = \sum_{n_k=0}^{\infty} e^{-\beta(n_k+1/2)\hbar\omega_k} = \frac{e^{-\beta\hbar\omega_k/2}}{1 - e^{-\beta\hbar\omega_k}}.$$

Because the modes are independent, the partition function of the whole system is a product rather than a sum:

$$Z = \prod_{k\neq 0} Z_k, \qquad \ln Z = \sum_{k\neq 0} \ln Z_k.$$

Thus, the total energy is

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= \underbrace{\frac{1}{2}\sum_{k\neq 0}\hbar\omega_k}_{E_{\text{zp}}} + \sum_{k\neq 0}\frac{\hbar\omega_k}{e^{\beta\hbar\omega_k} - 1}. \end{aligned}$$

The zero-point energy E_{zp} does not depend on T.

At low temperatures ($k_B T \ll \hbar\omega_0$), only low-frequency modes contribute significantly. Using $\omega_k \simeq \omega_0|k|$ and $\Delta k = 2\pi/N$, we obtain

$$\begin{aligned} E_{\text{th}} &\simeq \sum_{k\neq 0}\frac{\hbar\omega_0|k|}{e^{\beta\hbar\omega_0|k|} - 1} \\ &\simeq \frac{N}{2\pi}\int_{-\pi}^{\pi}\frac{\hbar\omega_0|k|}{e^{\beta\hbar\omega_0|k|} - 1}\,dk \\ &\simeq \frac{N}{\pi}\int_0^{\infty}\frac{\hbar\omega_0 k}{e^{\beta\hbar\omega_0 k} - 1}\,dk. \end{aligned}$$

Let

$$x = \beta\hbar\omega_0 k, \qquad dk = \frac{k_B T}{\hbar\omega_0}\,dx.$$

Then

$$\begin{aligned} E_{\text{th}} &= \frac{N(k_B T)^2}{\pi\hbar\omega_0}\int_0^{\infty}\frac{x}{e^x - 1}\,dx \\ &= \frac{N(k_B T)^2}{\pi\hbar\omega_0}\cdot\frac{\pi^2}{6} \\ &= \frac{\pi N k_B^2 T^2}{6\hbar\omega_0}. \end{aligned}$$

Therefore, the low-temperature heat capacity is

$$\boxed{C = \frac{\partial E}{\partial T} \simeq \frac{\pi N k_B^2}{3\hbar\omega_0}\, T}.$$

Thus, $C \propto T$ for a sufficiently large one-dimensional chain. The continuum approximation assumes that the temperature is still large compared with the spacing of the lowest nonzero-mode energies.

3 III

3.1 (9)

For the d-dimensional lattice, the equations of motion can be resolved into independent displacement components. For a simple hypercubic lattice, each component has the dispersion relation

$$\omega^2(\boldsymbol{k}) = 4\omega_0^2 \sum_{\alpha=1}^{d} \sin^2 \frac{k_\alpha}{2}.$$

In the long-wavelength limit ($|\boldsymbol{k}| \ll 1$),

$$\omega(\boldsymbol{k}) \simeq \omega_0 |\boldsymbol{k}|.$$

In d-dimensional wave-vector space, the number of modes in a spherical shell of radius k and thickness dk is proportional to

$$k^{d-1}\, dk.$$

Since $\omega \propto k$ at low frequencies, the density of states satisfies

$$\boxed{g(\omega) \propto \omega^{d-1}}.$$

The temperature-dependent part of the energy is

$$E_{\text{th}} \propto \int_0^\infty \frac{\hbar\omega\, g(\omega)}{e^{\beta\hbar\omega} - 1}\, d\omega \propto \int_0^\infty \frac{\hbar\omega^d}{e^{\beta\hbar\omega} - 1}\, d\omega.$$

Let $x = \beta\hbar\omega$. Then

$$E_{\text{th}} \propto T^{d+1} \int_0^\infty \frac{x^d}{e^x - 1}\, dx.$$

The integral is independent of temperature, so

$$E_{\text{th}} \propto T^{d+1}.$$

Consequently,

$$\boxed{C \propto T^d}.$$

In particular, the low-temperature heat capacity is proportional to T, T^2, and T^3 in one, two, and three dimensions, respectively. The d independent displacement components affect only the overall coefficient, not the power of T.